Lesson Notes

MODULE 6 | LESSON 1

THE BALANCE SHEET: LEVERAGE AND DEFAULT

Reading Time	60 minutes
Prior Knowledge	Call option payoff, Put option payoff
Keywords	Balance sheet, Fundamental accounting equation, Assets, Liabilities, Equity, Home equity loan (HEL), Leverage, Equity multiplier, Recourse / Non-recourse

In the last module, we looked more in depth at the securitization process and saw that it was rife with challenges, such as information asymmetries, moral hazards, and model risk. These in turn led to a ramp up in leverage in many corners of the economy, from the borrowers to the banks and investors. And finally, this resulted in what we now call the Great Financial Crisis.

In this section, we take up the subject of leverage again, from the perspective of the balance sheet. We describe the personal balance sheet and the corporate balance sheet, and we discuss leverage in the context of the balance sheet and certain financial products that enable higher leverage. We relate the mortgage once again (but in more detail) to options payoffs as we did in the last lesson. Specifically, we look at how the options model can help us understand the dynamics of the defaults that preceded the GFC.

1. Home Equity

In the last lesson, we discussed some important correlations involving the real estate market, e.g., between real estate prices and the stock market, and between default and leverage. Specifically, we saw how the credit-price effect describes the relationship between real estate prices and the stock market.

In this lesson, we discuss more specifically how real estate prices interact with equity, in a second sense of the word equity. Before, we talked about real estate prices and the stock market, but equity can also refer to the capital invested in a company in the form of stock. That is, the stock market is where the equity in a public company is traded. The equity in a company (which is tracked on the balance sheet) usually takes the form of stock; for this reason, investing in stock is also called investing in “equities.” Become very familiar with these two distinct but tightly interconnected meanings of the term “equity” as they will come up again and again. For this lesson, we mostly mean equity as the capital investment in a company, also known as the net worth or capital, which sits on the balance sheet.

The most important thing to know about the balance sheet is the **fundamental accounting equation**:

Assets = Liabilities + Equity

In essence, this equation shows you what is being balanced on the balance sheet. This equation is perhaps the most fundamental idea in corporate finance, and many important implications follow from it as we will see.

In the first “required reading” video "Introduction to Balance Sheets", you learn how individual people can also have a balance sheet, and this implies a personal net worth: the equity amount on their personal balance sheet. The second required video "More on Balance Sheets and Equity" further demonstrates how a personal balance sheet is impacted by the house price and mortgage.

For many homeowners, their home is their largest asset. And for many of these same homeowners, their mortgage is their main liability. If their home is worth *less* than they owe on their mortgage, they have “*negative equity*.” (They are also considered “*under water*” on their mortgage, and they are at risk of defaulting on their mortgage loan.) But if their home is worth *more* than they owe on the mortgage, we say they have “*equity in their home*.” Such a mortgagor may want to turn this equity in their home into cash in their bank account, which they can spend. As you learn about in the third required reading video "Home Equity Loans", a home equity loan (HEL) can accomplish this.

With these arguments, you are seeing the fundamental accounting equation

$$\text{Assets} = \text{Liabilities} + \text{Equity}$$

in action, and you should be able to work out the answer to the following problem.

Problem. *Say the only assets on your personal balance sheet (as described in the Khan Academy videos) are your home (Euro 495,000) and cash (Euro 5,000), and your only liabilities are your mortgage (Euro 400,000) and your home equity loan (Euro 90,000). How much equity do you have on your personal balance sheet*?

a) Euro 10,000

b) Euro 105,000

c) Euro 195,000

d) Euro 505,000

Answer. According to the problem statement, you have two assets (home and cash) worth a total of Euro 500,000:

$$\text{Assets} = 495,000 + 5,000 = 500,000$$

Your liabilities are the Euro 400,000 mortgage debt and the Euro 90,000 HEL, so

$$\text{Liabilities} = 400,000 + 90,000 = 490,000.$$

The fundamental accounting equation can be used to determine the equity amount

$$\text{Equity} = \text{Assets} - \text{Liabilities} = 500,000 - 490,000 = 10,000.$$

Hence, the answer is (a).

The more easily you can manipulate this simple equation, the better prepared you will be moving forward. In real life, in particular when the equation is applied to corporations, their balance sheets can be quite complex, consisting of many assets and many liabilities. However, it is important to be aware of the nuances we just covered and be able to use them to handle even complex balance sheets.

2. Corporate Equity

In section 1, we reviewed the simple personal balance sheet. We are ready now to look at a corporate balance sheet, which is structured in exactly the same way (Assets = Liabilities + Equity), just with different entries and somewhat larger amounts. The fourth required reading "The Balance Sheet" discusses the corporate balance sheet in general and also introduces a typical balance sheet for a bank. Banks are a very special type of corporate entity because of their centrality to the financial system, so it makes sense to have a more informed understanding of their balance sheet. Pay special attention to

1. The aspects of a bank's balance sheet that drive profitability;
2. The priority of stockholders in relation to creditors;
3. What happens when equity turns negative;
4. The typical assets and liabilities on a bank's balance sheet.

Example. *Consider bank X. Indicate if the following elements are to be included in assets, liabilities, or equity:*

1. *Cash*;
2. *Loans*;
3. *Deposits*;
4. *Bank Capital*
5. *Capital Reserves*
6. *Reserves for Bad Loans*

Answer.

1. Cash is clearly an asset. In fact, banks, due to their business and regulatory norms, must have a minimum amount of liquid cash available.
2. Loans are also an asset. The (primary) business of banks is that of lending money and earning interest. Therefore, loans can be compared to the plant and/or machinery for an industrial company. Hence, loans are assets.

3. Deposits are liabilities. This should be intuitive because banks use these liabilities to generate more income by using these deposits to finance loans for individuals, corporations, etc. Banks will be able to leverage this additional capital to make the extra income that they might have otherwise earned through the use of their own capital.
4. Bank capital is part of equity. The names should give it away, but we should be able to recognize that bank capital comprises funds raised by selling equity or financing that comes from retained earnings, i.e. part of the profits, that the bank does not pay out to its shareholders.
5. Capital reserves. If a company issue shares at a premium, that is the shares are sold for a higher price than their face value, the additional money generated is allocated to a share premium account which is one type of capital reserves. Another example, in the case of a manufacturing company more than a bank, is a scenario in which a company has a piece of machinery worth, say USD 100,000 and the company sells it for USD 140,000. The profit of USD 40,000 from this transaction is transferred to the company capital reserves. This type of reserves act as additional equity and are recorded as equity.
6. Reserves for bad loans. They are an allowance for doubtful loans that act as a contra account against the total loans. In other words, the bank expects that not all loans will be repaid back. The bank acknowledges an expense and creates an account in assets.

NOTE. Pay close attention to how reserves are presented because only in one instance they are to be included among the assets of the bank and in the other they will be automatically part of equity.

Example. *Assume that Bank XYZ has GBP 5,000,000 in cash, securities worth GBP 60,000,000, and loans totaling GBP 290,000,000. Furthermore, it has GBP 260,000,000 in deposits, and it also has borrowings for an amount of GBP 74,000,000. Finally, it has GBP 10,000,000 in reserves for bad loans. What is Bank XYZ's equity or capital*?

Answer. The problem is not really more difficult than the one we solved in Section 1 of this lesson. It all comes down to attributing each entry to the correct category of assets and liabilities and using the fundamental equation of accounting.

Now, the assets are cash, securities, and loans. As explained reserves for bad loans are also an asset

Liabilities consist of deposits and borrowings.

Hence,

$$\text{Assets} = 5,000,000 + 60,000,000 + 290,000,000 + 10,000,000 = \text{GBP } 365,000,000.$$

Also,

$$\text{Liabilities} = 260,000,000 + 74,000,000 = \text{GBP } 334,000,000.$$

Hence,

$$\text{Equity} = \text{Assets} - \text{Liabilities} = 365,000,000 - 334,000,000 = \text{GB } 31,000,000.$$

3. Measuring Leverage with the Equity Multiplier

Now that we know about the balance sheet, we can discuss the concept of leverage in a new way. Before, leverage described how we can buy more than we can afford, for example, by buying securities "on margin" or buying a home using a mortgage. We can actually quantify leverage using the main components of the balance sheet in several ways. One type of leverage ratio is called the equity multiplier[1]:

$$\text{Equity multiplier} = \frac{\text{Assets}}{\text{Equity}}.$$

Since Assets = Liabilities + Equity, you can substitute the right-hand side of this equation for assets in the above equation, as follows:

$$\text{Equity multiplier} = \frac{\text{Liabilities} + \text{Equity}}{\text{Equity}}.$$

This second formula makes it even clearer that leverage increases with liabilities. The more debt (liabilities) a person or a company has, the more leverage they have or the more "leveraged" they are.

Example. *Let's use our previous example of the homeowner: Assets = Euro 500,000, Equity = Euro 10,000. Then*

$$\text{Equity multiplier} = \frac{500,000}{10,000} = 50x.$$

Observe the "x" notation: we generally say an entity is "50x" leveraged when the equity multiplier is calculated to be 50.

If we had used the second formula, then Liabilities = Euro 490,000, and

$$\text{Equity multiplier} = \frac{490,000 + 10,000}{10,000} = 50x.$$

Obviously, we get the same answer either way, but the second way shows that the liabilities drive the leverage. (By the way, 50x would be considered highly leveraged.)

Example. Now, consider the same individual's circumstances, except assume the following:*

There is no mortgage and no HEL, so no liabilities. (Assume now that the homeowner never borrowed money to buy the house in the first place and never took out a HEL). Assume that the house is worth the same amount and the individual has the same amount of cash.

Based on these assumptions, assets are still Euro 600,000 (= Euro 400,000 + Euro 200,000). Since Assets = Liabilities + Equity, and there is no debt or other liabilities, we find that

$$\text{Equity} = \text{Assets} - \text{Liabilities} = 600,000 - 0 = 600,000.$$

This yields

$$\text{Equity multiplier} = \frac{\text{Assets}}{\text{Equity}} = \frac{600,000}{600,000} = 1x.$$

When the equity multiplier is 1x, we say that the entity is not leveraged.

As we discussed in the last lesson, higher leverage usually means higher risk. One way of seeing that higher leverage means higher risk is by noting that higher leverage implies lower equity, and equity is the buffer between assets and liabilities: when liabilities are greater than assets, the negative equity means that a borrower might have trouble making payments that are owed.

4. Modeling Default with Options

Now that you are comfortable with the idea of equity, how equity can change, and how equity can be positive or negative, you are ready for the next required reading (Foote and Willen) for this lesson, which uses options theory to explain why and when mortgage borrowers default. It may help you to review the earlier discussion (in Module 5, Lesson 2) about how a mortgage is like an option. A model that accurately describes or predicts probability of default can be very useful from a credit risk perspective and from a valuation perspective: if you can be more confident about the level of default in a mortgage pool, you can be more confident about the appropriate price to pay for the MBS with that mortgage pool as the underlying.

You will see in the new reading that a borrower's mortgage can be seen as a long call option on the value of the house: as with any long option, there is potential for significant payout (say, if the house price rises), but the potential loss is relatively minimal (if the house price declines, the borrower only loses their equity).

A mortgage can also be seen as a long put with a strike at the mortgage amount, combined with the underlying security. If the house price increases, the borrower profits from the increased equity in the home, but if the house price declines below the mortgage amount, the borrower can sell the underlying house to the lender by defaulting. You may want to review the option payoff diagrams in Module 4, Lesson 1 to remind yourself of these dynamics. The application of options theory beyond calls and puts on stocks should become very natural to you as you analyze more exotic financial products and relationships.

Pay special attention to the development of the model to explain mortgage defaults, from these simple options analogies of the frictionless option model (FOM) to the double trigger model (DTM) and the more sophisticated combinations of the DTM and FOM models. Also, note how the assumptions involved change from model to model, and ask yourself if the assumptions are realistic. For example, what is the role of negative equity in each model? Does negative equity alone cause mortgagors to default? For example, which models assume that the mortgagor can borrow funds to make the mortgage payment? The evaluation of a model's assumptions is critical to your decision making about which models make sense to use in which situations.

5. Recourse vs. Non-Recourse

While we mentioned net worth is the equity on a personal balance sheet, most of our discussion of equity has focused on home equity: the equity in a house that is mortgaged, whether that equity is positive or negative. Obviously, “home equity” and the equity on a personal balance sheet (net worth) are related: significant home equity may translate into significant personal equity, or at the very least, it will offset some of the other liabilities on the personal balance sheet to increase the personal equity somewhat. But while these two equities are related, they are not the same.

As you will read about in this required reading (Harris and Meir), most mortgages in the United States are “non-recourse,” meaning that the lender can take control of the collateral (the house in the case of a mortgage), but the lender cannot make any additional claims on the borrower, even if the proceeds from the sale of the house do not cover the mortgage amount: we say the lender “does not have recourse” to the borrower with a non-recourse mortgage. In Europe, on the other hand, mortgage lenders generally do have recourse, meaning they can take and sell the house and then continue to seek repayment from the borrower. The paper infers some interesting ideological differences between European nations and the United States. More important for our purposes, however, is the reappearance of moral hazards and information asymmetries that follow from both types of mortgages. Pay attention in this reading to how recourse mortgages allocate the risks of default very differently compared to non-recourse mortgages.

6. Conclusion

In this lesson, we learned the three main components that balance the balance sheet, be it a personal balance sheet or a corporate balance sheet like that of a bank. We calculated the leverage ratio using these components and saw how debt and leverage are very intertwined. We looked at how models can be constructed from previous models, such as options theory, to explain complicated phenomena such as default behavior, which can be very useful in the valuation of MBSs. We also took note of how moral hazards and information asymmetry seem to lurk in so many corners of financial markets and how government policies respond to these by, for example, allowing or prohibiting recourse mortgages. We are seeing that financial engineering must take into consideration a wide array of behavioral and regulatory issues in order to be practical.

So far, we have largely focused on how to finance home ownership; in the next lesson, we talk about renting. Specifically, we discuss when it is better to rent vs. to own from a financial perspective. We also discuss real estate development, without which there would be no housing to rent or own. Thus, development must also be financed, so we discuss the ways that developers finance their construction, etc. This discussion necessarily brings us back to the balance sheet, as equity and debt are the main sources of financing almost anything.

Footnotes

1. It is also worth noting that this “balance sheet leverage” can be related to the debt-to-income (DTI) ratio that we discussed in the context of a borrower’s credit risk--but don’t confuse them. Balance sheet leverage is based on the components of the balance sheet; DTI uses income and debt service payments, which are components of an income statement. Both ratios capture the leverage and therefore the credit risk of an entity (a company or person).

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